

Auditing Weather-Feature Reliance in Detrended Crop-Yield Models

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Abstract—Interpretable machine learning cannot support substantive claims about below-trend crop-yield events unless its fitted predictive model has adequate out-of-sample fidelity. We audit this condition in a U.S. crop-state-year panel with 1257 rows for Barley, Canola, Oats, and Wheat from 1990–2025. Trends and residual scales are fit on training years only; selection uses 2012–2015 validation, and 2016–2025 remains locked until final evaluation. The validation-selected ExtraTrees weather only model has final $R^2 = -0.014$, RMSE 0.669 t ha^{-1} , and paired ΔRMSE versus a zero-residual baseline of -0.005 with upper 95% bound 0.009 . Its fidelity gate fails. We therefore retain SHAP, LIME, permutation, ablation, and response-curve results only as descriptions of a fitted function, not evidence that weather caused observed events. The contribution is a reproducible stop-gate protocol with paired year-block uncertainty, tail checks, robustness checks, and an explicit warning that full-series detrending is retrospective only.

Index Terms—crop yield, explainable machine learning, temporal validation, negative results, reproducibility

I. INTRODUCTION

Machine-learning yield forecasts are increasingly built from weather, remote sensing, management, and crop-model variables. Studies report useful predictive systems across large and small panels, diverse model families, and hybrid formulations [1]–[7]. Yet a plausible feature attribution is not causal evidence, and agreement between explanation methods cannot compensate for a model that does not predict the future target reliably. This distinction is especially important for below-trend yield events, where a compelling explanation of an individual fitted prediction can be mistaken for an explanation of the observed loss.

Weather extremes and climate variability are materially associated with crop-yield variation at several spatial scales [8]–[16]. Those findings motivate careful predictive evaluation; they do not license an event-level attribution from this restricted panel. We ask a narrower question: does a leakage-safe residual model have enough future-period fidelity and stability to permit substantive weather-feature-reliance interpretation? The answer is evaluated before XAI is interpreted.

The paper makes three contributions. First, it uses train-only target construction and validation-then-locked-test selection. Second, it specifies a quantitative fidelity gate with same-task baselines, paired year-block intervals, tail metrics, and pre-specified robustness checks. Third, it contrasts a prospective target with a retrospective full-series detrending diagnostic. A gate failure is a negative methodological result: it stops causal or event-driver claims, rather than choosing the most attractive explanation.

II. RELATED WORK

A. Yield Modeling and Detrending

Detrending separates a local historical yield trajectory from residual variation, but the result can depend on the detrending procedure and information available at the evaluation date [17], [18]. We therefore fit each crop-state trend and residual scale on training observations only. Ridge regression, Random Forest, and ExtraTrees provide deliberately distinct linear and tree-based model families [19]–[21]; the implementations use scikit-learn [22]. This is a finite pre-specified comparison, not a search over the locked test period.

B. Explanation Methods and Their Boundary

SHAP and LIME explain a model output under their respective approximation and reference choices [23], [24]. Permutation-style analyses, model reliance, accumulated local effects, and conditional importance likewise describe feature dependence or behavior of a fitted predictor [25]–[27]. They can be useful diagnostic artifacts. However, none turns failed prospective predictive fidelity into evidence about an observed yield anomaly. The gate below formalizes this boundary.

C. Evaluation and Reproducibility

Random splits can be inappropriate when temporal or hierarchical structure is present [28]. Leakage can create seemingly strong scientific machine-learning results while invalidating the intended forecast claim [29]. Reproducible ML also requires explicit code, data paths, settings, and numerical evidence [30]. Accordingly, every manuscript number is generated from an auditable run: configuration, prediction vectors, bootstrap draws, citation check, and final PDF are all versioned outputs.

III. DATA AND PROTOCOL

A. Panel, Features, and Target

The unit is a crop-state-year row. The processed panel contains 1257 observations from 1990–2025 for Barley, Canola, Oats, and Wheat across 12 U.S. states. Yield in t ha^{-1} comes from USDA NASS Quick Stats [31]; daily maximum/minimum temperature, precipitation, and shortwave radiation come from NASA POWER [32]. The primary design has 35 full-season weather features. A separate sensitivity design adds early-, middle-, and late-season proxies without redefining the target or locked population.

For a test interval, each crop-state linear trend and residual scale is estimated using only its associated training rows. A

TABLE I
DATA-FLOW AND FINAL-TEST ELIGIBILITY RECONSTRUCTED FROM THE RAW INPUTS.

Stage	Rows	Rule
raw_yield_file	1291	source table supplied in repository
analysis_year_filter	1257	exclude 34 raw 1989 rows because analysis starts in 1990
processed_model_frame	1257	exact reconstruction from raw yields plus NASA POWER daily input
final_test_eligible	333	2016-2025 rows with at least three prior series observations

row is eligible only with at least three training observations. The validation period contains 140 eligible rows after 4 pre-specified exclusions. The locked test starts with 343 candidates, excludes 10, and evaluates 333 rows. Future yields cannot change a previously constructed evaluation target; this property has a unit test and row-level audit. The learned target is the raw train-only residual in t ha^{-1} ; the standardized residual $z = r/\hat{\sigma}_{\text{train}}$ defines below-trend events only. This distinction prevents a pooled raw-error score from being misread as a standardized event-recovery metric. Min-history rules of 3, 5, 8, and 10 years are reported as sensitivity analyses because both eligibility and the fitted residual scale can change with available history.

B. Selection and Baselines

Ridge $\alpha \in \{1, 10\}$ and Random Forest/ExtraTrees with 160 trees and $\text{min_samples_leaf} \in \{1, 2\}$ are evaluated on metadata-only, weather-only, and full representations. Stochastic fits use fixed seeds $\{7, 17, 27, 37, 47\}$; numeric imputation/scaling, categorical encoding, and model fitting occur within the training fold. Mean per-row seed-aggregated validation RMSE chooses the configuration, with lexicographic tie-breaking and no locked-test access. Seed-level RMSE standard deviations are reported so tiny differences are not presented as deterministic wins.

The final audit compares learned predictions on identical rows to zero residual, training-mean residual, and crop training-mean residual baselines. Prediction vectors are hashed and pairwise maximum absolute differences are retained, so any nearly coincident baselines are demonstrated rather than assumed. We use 2,000 deterministic calendar-year block bootstrap replicates and percentile 95% intervals. An invalid replicate fails the run; none are silently dropped. This block-resampling choice follows the dependent-observation motivation of the stationary-bootstrap literature [33]; state-year and crop-state cluster schemes are reported as pre-specified sensitivity analyses.

C. Pre-specified Fidelity Gate

For a paired comparison, model-minus-baseline error passes only when the upper 95% year-block interval is below zero. Gate A is predictive fidelity of the validation-locked choice; Gate B is incremental weather value relative to context. Gate B never triggers re-selection. The primary operational tail is $z < -1$; $z < -1.5$ and $z < -2$ are sensitivity analyses because their smaller samples reduce power. Sign accuracy on a negative-only subset is reported only as a diagnostic, not a gate component. Instead, rank recovery requires a positive lower bootstrap bound and a within-year permutation

TABLE II
FIXED GATE POLICY. GATE A AND GATE B ARE INDEPENDENT; MACHINE-READABLE VALUES ARE IN `configs/fidelity/gate.yaml`.

Component	Pre-specified pass rule
Overall	upper paired 95% CI of selected-minus-zero RMSE < 0
Tail $z < -1$, -1.5 , -2	RMSE and MAE CI upper < 0 ; sign ≥ 0.50 , $\rho \geq 0$, top- $k \geq 0.10$
Rolling origin	at least 3/4 folds improve RMSE over zero
Incremental weather	upper paired 95% CI of full-minus-metadata RMSE < 0

TABLE III
VALIDATION SELECTION ON 2012–2015. RMSE IS COMPUTED AFTER FIXED-SEED PREDICTION AGGREGATION; SEED SD IS THE ACROSS-SEED RMSE STANDARD DEVIATION.

Config	Features	n	Seeds	RMSE	Seed SD	Selected
extra_trees_leaf_1	weather_only	140	5	0.383	0.003	Yes
extra_trees_leaf_2	weather_only	140	5	0.384	0.001	No
random_forest_leaf_1	weather_only	140	5	0.386	0.003	No
random_forest_leaf_2	weather_only	140	5	0.387	0.002	No
extra_trees_leaf_2	full	140	5	0.388	0.003	No
random_forest_leaf_1	full	140	5	0.389	0.002	No

$p < .05$. Top- k recovery reports random expectation k/n , lift, hypergeometric probability, permutation p , and a year-block lift interval. Thus a severe-event row cannot pass simply by overlapping one arbitrary event. The fixed policy is in Table II.

Algorithm 1: Locked fidelity audit

Input: panel, fixed split/configuration, seed list, gate configuration.

1. Fit detrending and preprocessing on each training period only.
2. Select one configuration on validation RMSE; lock it.
3. Refit the locked choice through 2015 and predict 2016–2025.
4. Compare same-row baselines with paired year-block bootstrap intervals.
5. Evaluate null-aware primary-tail rank/top- k recovery and sensitivity tails.
6. Return Gate A and Gate B separately; stop substantive event claims when Gate A fails and weather-specific claims unless both gates pass.

IV. RESULTS

A. Validation and Locked Test

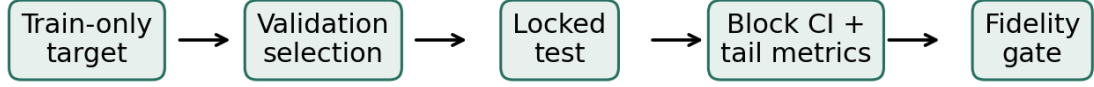
Table III shows leading validation configurations. The selected model is ExtraTrees with weather only features. The complete grid, seed-level metrics, tie rule, and locked-test access flag are supplied in the artifact. A non-selected ExtraTrees full model can have a small final point estimate without changing the decision: it was not validation-selected, and the locked test is never used to switch configurations.

The selected weather-only ExtraTrees configuration ranks first in 3/5 stochastic validation seeds. Its advantage is small relative to several seed standard deviations, so the selected row is a reproducible protocol choice rather than a claim of decisive superiority. The one-standard-error flag and every seed-level score are retained; no final-test statistic changes this selection.

On the locked test, the selected model has $R^2 = -0.014$ and RMSE 0.669 t ha^{-1} . Its paired RMSE difference from zero crosses zero, so G1 fails. Table V includes all same-task naive baselines and the selected configuration across feature families. Figure 2 deliberately separates selected-model predictive fidelity from the distinct full-versus-metadata incremental-weather comparison; neither is a post-hoc model-selection mechanism.

B. Event-Detection Diagnostics

Event detection is evaluated over all 333 locked rows, not only a tail preselected by the observed outcome. The score is



PASS: interpret; FAIL: stop

Fig. 1. Prospective audit workflow. Selection is complete before the locked 2016–2025 test; the final fidelity gate controls whether interpretation may proceed.

TABLE IV
VALIDATION STABILITY OF THE LEADING CONFIGURATIONS. THE RANK PROBABILITY IS ACROSS FIXED STOCHASTIC SEEDS; 1-SE DENOTES INCLUSION UNDER THE PRE-SPECIFIED ONE-STANDARD-ERROR DIAGNOSTIC, NOT A POST-HOC REPLACEMENT.

Config	Features	Mean RMSE	SD	P(rank=1)	1-SE
extra_trees_leaf_1	weather_only	0.384	0.003	0.60	Yes
extra_trees_leaf_2	weather_only	0.385	0.001	0.20	No
random_forest_leaf_1	weather_only	0.387	0.003	0.20	No
random_forest_leaf_2	weather_only	0.387	0.002	0.00	No
extra_trees_leaf_2	full	0.389	0.003	0.00	No
random_forest_leaf_1	full	0.390	0.002	0.00	No

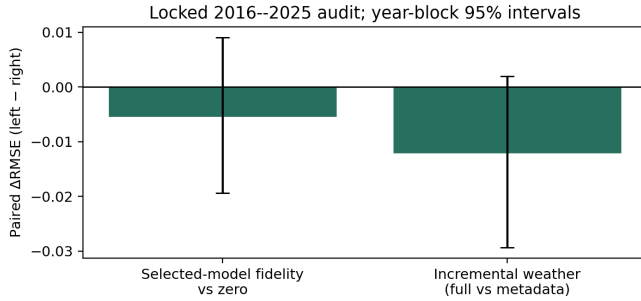


Fig. 2. Two distinct locked-test comparisons. Negative values favor the left model; bars are paired 95% year-block intervals.

negative predicted residual and the score threshold is fixed at zero before final evaluation. Table VI reports PR-AUC, ROC-AUC, balanced accuracy, precision, recall, and F1. These are diagnostics rather than Gate A components: they describe discrimination, but they do not override the paired-error and null-aware rank failures. In particular, an always-negative predictor cannot gain a gate pass through the negative-only sign rate used in older drafts.

C. Below-Trend Gate and Robustness

The broad below-trend subset contains 73 locked-test rows. Table VII separates error, rank-null, and chance-adjusted event-recovery components. The primary $z < -1$ top-10 lift is below random expectation and its permutation result is not significant; rank uncertainty also crosses zero. Gate A therefore fails without relying on a directional-sign statistic. Neither local explanations nor global/group diagnostics are

evidence of weather causes or event drivers in this primary analysis.

Table IX gives exact temporal pass counts and summary ranges for the pre-specified stage-feature, crop, spatial, and target-protocol checks. These checks do not reverse the central decision; they show its stability boundary rather than optimize around it.

D. Target, Cluster, and Model-Family Sensitivity

The primary rule requires at least three earlier observations for a crop–state trend. Increasing this rule to eight years changes the final population from 333 to 313 rows and changes the paired overall interval. That is not used to replace the pre-specified primary result: it demonstrates that eligibility and trend scale are consequential design choices. We therefore report it as a limitation rather than select the favorable sensitivity. State-year and crop–state cluster resampling also retain an interval crossing zero under the primary rule. Finally, HistGradientBoosting and ElasticNet are added as pre-specified tabular comparators; their locked-test point estimates are diagnostic, not post-hoc replacements for the validation-selected configuration. Table X preserves the exact population, interval, and status for these checks.

E. Audit Traceability and Interpretation Limits

Table XI maps each numerical claim family to a concrete artifact. This is intentionally more than a code-release statement. E1 records the temporal split and train-only target construction; E2 records the entire validation grid, fixed seed outcomes, and tie rule; E3 stores every sampled calendar-year block; and E4 records every tail prediction, metric, and paired interval. The remaining checks preserve the temporal, feature, crop, spatial, target-construction, and baseline evidence required to reproduce the stop decision. Consequently, a reader can distinguish an empirical model result from a narrative inference about weather.

Several limits should shape interpretation. State-representative weather coordinates can miss within-state exposure; the weather covariates are correlated; and this panel lacks soil, irrigation, cultivar, management, and phenological measurements. The response variable is a detrended state aggregate rather than a farm-level loss. The observed gate

TABLE V

LOCKED 2016–2025 SAME-TASK AUDIT. LEARNED-MODEL PREDICTIONS ARE SEED AGGREGATED. ΔRMSE IS MODEL MINUS ZERO-RESIDUAL RMSE WITH PAIRED 95% YEAR-BLOCK INTERVALS; RMSE IS t ha^{-1} .

Model	Features	n	Seeds	R^2	RMSE	ΔRMSE vs zero [95% CI]
Crop train mean	baseline	333	1	-0.031	0.674	0.000 [-0.000, 0.000]
Train mean	baseline	333	1	-0.031	0.674	0.000 [0.000, 0.000]
Zero residual	baseline	333	1	-0.031	0.674	0.000
ExtraTrees	full	333	5	0.006	0.662	-0.012 [-0.029, 0.002]
ExtraTrees	metadata_only	333	5	-0.031	0.674	0.000 [-0.000, 0.000]
ExtraTrees	weather_only	333	5	-0.014	0.669	-0.005 [-0.019, 0.009]

TABLE VI

LOCKED-TEST EVENT-DETECTION DIAGNOSTICS ON ALL ROWS. THESE VALUES ARE NOT TUNED ON THE FINAL TEST AND DO NOT REPLACE GATE A COMPONENTS.

Tail	Prevalence	PR-AUC	ROC-AUC	Bal. acc.	F1	Role
$z < -1$	0.219	0.338	0.641	0.589	0.383	DIAGNOSTIC
$z < -1.5$	0.156	0.294	0.672	0.611	0.330	DIAGNOSTIC
$z < -2$	0.108	0.217	0.672	0.598	0.244	DIAGNOSTIC

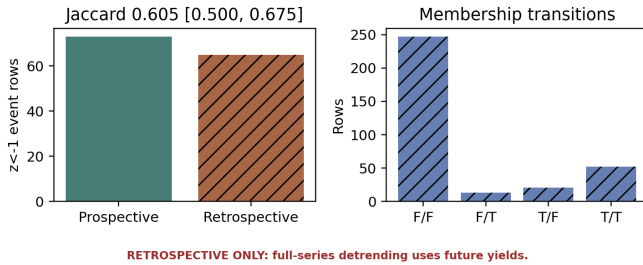


Fig. 3. **RETROSPECTIVE ONLY.** Full-series detrending uses future yields. Bars show exact event counts and the transition panel shows prospective/retrospective membership; this is not prospective forecast evidence.

failure may therefore reflect a mismatch among target, resolution, features, and model family, not an absence of physical weather effects. Conversely, these limitations do not justify relaxing the gate after observing the locked test. They identify inputs for a future, separately specified study: richer spatial covariates, prospectively defined phenology windows, and an independent evaluation period.

F. Retrospective Target Sensitivity

Full-series detrending uses later yields to construct earlier residuals. It is therefore a retrospective diagnostic, not a prospective target. Relative to the train-only construction, the $z < -1$ final-period set has Jaccard overlap 0.605 with year-block 95% interval [0.500, 0.675]; 21 prospective events are not retrospective events and 13 retrospective events are not prospective events (Fig. 3). A coherent retrospective explanation would still not repair the prospective fidelity failure.

V. DISCUSSION AND CONCLUSION

The result is informative because the model was selected without final-test access, compared with same-task baselines,

evaluated with paired uncertainty, stress-tested on negative tails, and checked across temporal, feature, crop, spatial, and target protocol settings. The evidence does not support substantive weather-feature claims for observed below-trend events in this crop–state panel. It does not imply that weather effects are absent; it limits what the present target, features, and model families can establish.

The reusable lesson is procedural. Lock selection before testing; distinguish G1 predictive fidelity from G2 incremental feature value; require paired uncertainty and tail recovery; expose baseline vectors; and stop interpretation when any pre-specified component fails. The anonymized artifact contains raw-input reconstruction evidence, configuration files, row-level predictions, bootstrap draws, reference/citation audits, generated figures/tables, and one-command reproduction. USDA NASS and NASA POWER remain subject to their source terms.

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TABLE VII

TAIL COMPONENTS ON TRAIN-ONLY TARGETS. ONLY $z < -1$ IS PRIMARY; SEVERE TAILS ARE SENSITIVITY ANALYSES. A STATUS IS A DEFINED COMPONENT RESULT, NOT AN ARTIFACT-VERIFICATION LABEL.

Role	Tail	n	Δ RMSE [CI]	Δ MAE [CI]	Rank ρ [CI]; p	Top-10 lift; p	Status
Primary	$z < -1$	73	-0.043 [-0.074, 0.007]	-0.054 [-0.096, 0.002]	0.180 [-0.035, 0.373]; p=0.362	0.73; p=0.793	FAIL
Sensitivity	$z < -1.5$	52	-0.056 [-0.083, -0.002]	-0.073 [-0.109, -0.007]	0.060 [-0.151, 0.416]; p=0.511	0.52; p=0.935	FAIL
Sensitivity	$z < -2$	36	-0.056 [-0.090, 0.014]	-0.076 [-0.118, 0.015]	-0.051 [-0.306, 0.497]; p=0.797	0.36; p=0.985	FAIL

TABLE VIII

GATE A/B DECISION COMPONENTS. GATE A CONTROLS SUBSTANTIVE OBSERVED-EVENT CLAIMS; GATE B IS ADDITIONALLY REQUIRED FOR WEATHER-SPECIFIC CLAIMS.

Gate	Component	Scope	Status
Gate A	overall paired RMSE	locked final test	FAIL
Gate A	primary-tail paired RMSE	$z < -1$	FAIL
Gate A	primary-tail paired MAE	$z < -1$	FAIL
Gate A	rank recovery	$z < -1$	FAIL
Gate A	chance-adjusted top-k recovery	$z < -1$	FAIL
Gate B	extra_trees_leaf_1_weather_only_vs_metadata_only	locked final test	FAIL
Gate B	extra_trees_leaf_1_full_vs_metadata_only	locked final test	FAIL
Gate A	FINAL GATE A	$z < -1$	FAIL
Gate B	FINAL GATE B	weather-specific claim	FAIL

TABLE IX

ROBUSTNESS AND PROTOCOL SENSITIVITY. COMPLETE ROW-LEVEL OUTPUTS AND SETTINGS ARE INCLUDED IN THE AUDIT ARTIFACT.

Audit	Result	Boundary
Rolling origin	3/4 improve; R^2 -0.081 to 0.159	Temporal
Stage proxies	50 features; Δ RMSE -0.001	Sensitivity
Crop-specific	4 crops; R^2 -0.273 to 0.072	Crop
Leave-one-state-out	4/12 improve over zero	Spatial
Detrending	Jaccard 0.605; rank ρ 0.942	Protocol

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TABLE X
TARGET, CLUSTER, AND EXPANDED-MODEL SENSITIVITY. HISTORY 3 IS THE PRIMARY PRE-SPECIFIED ELIGIBILITY RULE; LATER HISTORY RULES ARE NOT USED TO CHANGE SELECTION OR THE PRIMARY CLAIM.

Analysis	n	RMSE/ Δ	95% interval	Status
History 3	333	0.669	[-0.019, 0.009]	FAIL
History 5	333	0.669	[-0.019, 0.009]	FAIL
History 8	313	0.530	[-0.041, -0.007]	PASS
History 10	313	0.530	[-0.041, -0.007]	PASS
year_block	10	-0.005	[-0.019, 0.009]	FAIL
state_year_block	120	-0.005	[-0.022, 0.010]	FAIL
crop_state_cluster	36	-0.005	[-0.033, 0.019]	FAIL
hist_gradient_boosting	333	0.675	locked point estimate	DIAGNOSTIC
elastic_net	333	0.674	locked point estimate	DIAGNOSTIC

TABLE XI
E1–E10 NUMERICAL TRACEABILITY. VERIFIED MEANS THE EVIDENCE ARTIFACT WAS REGENERATED AND CHECKED; IT DOES NOT MEAN A SCIENTIFIC GATE PASSED.

ID	Numerical trace	Status
E1	train-only target and split manifest	PASS
E2	validation seed metrics and tie rule	PASS
E3	2,000 paired year-block draws	PASS
E4	all fixed tail metrics and intervals	PASS
E5	four rolling-origin folds	PASS
E6	stage-feature sensitivity	PASS
E7	crop and leave-one-state-out checks	PASS
E8	retrospective detrending sensitivity	PASS
E9	baseline vectors, hashes, differences	PASS
E10	numbers generated from audit records	PASS

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